EXAMS

Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications Technology Department

Academic Year	2015 / 2016
Semester	First
Exam Date	18/11/2015

QUESTIONS FOR THE Mid Term EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiner:	Dr. Samir Kamal	**Time:**	75 Mints

Number of Pages: one | Number of Questions: **2** | Attempt all questions

Model-A

Question 1 [20 Points]

(a) In A/D conversion, in detail explain the coding process.

(b) Find the relationship between the frequencies, f, F & F_S, where these frequencies respectively are: frequency of analog signal, frequency of discrete-time signal and sampling frequency.

(c) For the signal $x(t) = \cos(300\pi t)$ to be sampled with sampling frequency F_S = 300Hz.

(i) Is F_S is suitable for sampling the given signal, $x(t)$, if it is suitable, find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.

(ii) Find the samples values of the first cycle in x(n), then plot $x(t)$ and x(n) on the same figure.

Question 2 [20 Points]

(a) Show that the DFT, X(K), of a sequence x(n) is periodic.

(b) For the following sequence: x(n) = {1, 0, 1, 0}:

(i) Using the direct method, DFT, find the frequency components of this sequence and check your results by calculating x(n) using the IFFT algorithm.

(ii) Using your results, verify Parseval's theorem.

(iii) If we use FFT algorithm instead of the DFT algorithm, what is the computational saving?

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Model-B

Question 1 [20 Points]

(a) In A/D conversion, in detail explain the quantization process.

(b) Given two signals: $x_1(t) = 5\cos(200\pi t)$ and $x_2(t) = 2\sin(500\pi t)$ to be sampled with sampling frequencies F_{S1} = 400Hz and F_{S2} = 1KHz respectively.

(i) Are the sampling frequencies given above are suitable for sampling the given signals, $x_1(t)$ and $x_2(t)$

(ii) If they are suitable, use them to find expressions for $x_1(n)$ and $x_2(n)$.

Question 2 [20 Points]

(a) Show that the DFT, X(K), of a sequence x(n) is periodic.

(b) For the following sequence: x(n) = {0, 1, 0, 1}:

(i) Using the direct method, DFT, find the frequency components of this sequence and check your results by calculating this frequency components using FFT algorithm.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.

G O O D L U C K

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QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (E501) Digital Signal Processing **Spec.:** 5th Elec. Eng. & Comm. Tech. Dep.

Examiners: Dr. Samir Kamal and Prof. Mostafa Afifi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

Question 1 **(20 points)**

a - In detail, what are the main characteristics of the continuous time sinusoidal signals?

b- Check the periodicity of the following sequences, showing their fundamental periods.

(i) y (t) = $7\cos(400\pi t)$

(ii) x (n) = $\sin(2n + \pi/4)$

(c) For the signal $x(t) = 2\cos(3000\pi t)$ to be sampled with sampling frequency F_S = 3 KHz.

(i) Is F_S suitable for sampling the given signal, $x(t)$,

(ii) If it is suitable, use it to find an expression for the corresponding sequence, x(n),

(iii) Calculate the fundamental period, N_P, of the obtained x(n),

(iv) Find the values of x(n) in the first cycle and

(v) Plot (showing the steps) the given $x(t)$ and the obtained x(n) on the same figure.

Question 2 **(25 points)**

a- In A/D conversion, in detail explain the sampling process.

b - For the following sequence: x(n) = {1, 1, 0, 1}:

(i) Using the FFT algorithm, find the frequency components of this sequence and check your results by calculating x(n) using the IDFT method.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.

c- The transfer function for an FIR filter is given by: $H(z) = 1 - 1.4z^{-1} + 0.9z^{-2}$. Design and realize the block diagram for this filter in the following cases:

(i) Transversal structure.

(ii) A two-stage lattice structure.

Question 3 **(15 points)**

a- In correlation operation define the following items: Correlation operation, correlation coefficient and the end effect problem.

b - Given two sets of samples $x_1(n)$ = {1, 2.5, 1} and $x_2(n)$ = {0, 2, 1.5, -1}.

(i) Find the cross- correlation between $x_1(n)$ and $x_2(n)$.

(ii) Plot the correlation function obtained in part (i) against the lag number, j.

(iii) Find the autocorrelation function of the sequence $x_1(n)$.

c- Find the output of an electrical system, y(t), with impulse response h(t) shown in Fig. 1a when the input signal, x(t) is the signal shown in Fig. 1b. Plot the obtained y(t) v.s t,

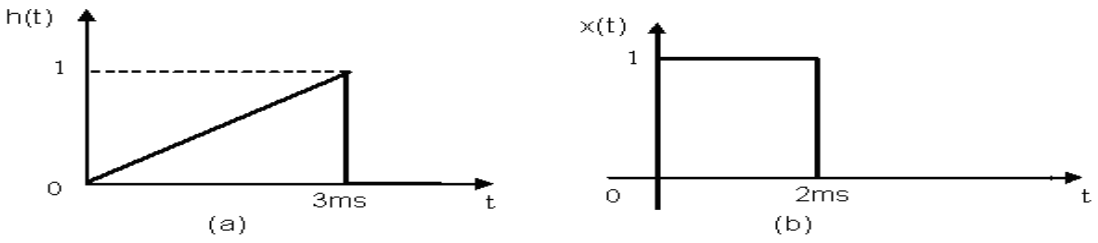


Fig. 1

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Question 4 **(20 points)**

a- For LTIS define: Causality, linearity and the difference equation.

b- By applying a unit impulse to the discrete-time system characterized by the following z-transfer function, find its impulse response, h(n), (calculate the first 4-values, assuming an initial condition of y(–1) = 0).

$$H(Z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} - 2z^{-2}}$$

c- Using the residue method, find the impulse response, h(n), of the system characterized by the following transfer function, $H(z)$:

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

For this transfer function:

(i) Plot its pole-zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system,
(v) Draw its filtering operation in a block diagram form and
(vi) Determine the computational and storage requirements.

Question 5 **(10 points)**

Check (✔) for the right statements and (X) for the wrong statements and correct the mistakes.

(1) In quantization process, when the quantization step (Δ) increases, the accuracy of the Quantizer increases. ( )
(2) The smaller the quantization step the longer will be the required codeword length. ( )
(3) The spectrum of a continuous periodic signal could be obtained using Fourier transform whereas the spectrum of discrete time signal could be obtained using the FFT algorithm.( )
(4) FFT is calculated for a signal that has any number of samples. ( )
(5) Correlation coefficients never exceed one. ( )
(6) We can calculate the normalized energy of any signal performing convolution at lag=1( )
(7) A discrete time system is said to be time invariant if its output is independent of the time the input is applied. ( )
(8) The impulse response of a discrete- time system may be defined as the z- transform of the system's transfer function H (z). ( )
(9) IIR requires more coefficients for sharp cut off filters than FIR. ( )
(10) For a given amplitude response specification more processing time and storage will be required for IIR filter implementation. ( )

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